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Next Generation Paraffin Inhibitor / Pour Point Depressant Screening: Automated High Throughput Experimentation Method

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Abstract

Paraffin deposition is a critical flow assurance challenge and chemical treatment with continuous Paraffin inhibitor injection has been the go-to approach for operators to preemptively tackle this challenge. Development of the right inhibitor chemistry and dosage is of critical importance and can be challenging. A smart, automated process for screening a high number of paraffin inhibitor formulations based on a Design of Experiments and using High Throughput Experimentation has been developed and is detailed in this work.

Paraffin inhibitor formulations comprising distinct base chemistries, boosters, and solvent packages were blended in various ratios to encompass a large chemical and formulation space with an automated blending station of the High Throughput Experimentation (HTE) setup. These formulations were then dosed into multiple crude oil samples from different regions of the world at constant dosage to represent a uniform and comparable test fluid matrix. The rheological behavior of paraffin clusters within dosed oil samples were then evaluated using HTE techniques to generate numerous performance data points. Furthermore, the efficiency of the top performing formulations was then cross validated using standard rheometer and deposition-based techniques like cold finger to gauge performance.

The four main pillars of HTE are automation, miniaturization, parallelization, and computational design. Each of these pillars contributes towards enhancing the current test method into a more accurate, agile, and quicker technique. Through automation, the overall accuracy of formulation blending as well as performance efficiency measurement is increased. With miniaturization, the resource consumption of raw materials and more importantly crude oil is considerably reduced. For this work, each formulation required less than 300 gl of oil per measurement. Parallelization resulted in completing the test evaluation at a rate of 600 evaluations every 4 hours. These results can then be analyzed through the computational design and analytics software to identify the top performing formulations along with predicting the optimum chemical formulation space which can lead to maximum performance efficiency. The selected formulations were subsequently blended in a larger scale and their efficacy was cross-checked with both rheological (rheometer) and deposition (cold finger and wax loop) based techniques against a benchmark paraffin inhibitor product, known to work successfully in many fields which exhibit wax instability related issues.

Through this process, several new paraffin inhibitor formulations were discovered that outperformed the benchmark blend.

Automated HTE technology provides a new dimension to paraffin inhibitor development work that can carry out the performance evaluation of numerous formulations at tremendous speed and minimum crude oil volume. With this technique, to the changing requirements of paraffin inhibitor and its dosage with varying crude oil composition or any operational change affecting overall paraffin stability are possible.

Introduction

Flow assurance challenges in oil production and transportation systems represent some of the most persistent and costly operational issues facing the petroleum industry today. Among these challenges, asphaltene deposition and paraffin wax precipitation stand out as particularly problematic phenomena that can severely impact production efficiency and system integrity. As the industry moves toward more challenging production environments—including deepwater, ultra-deepwater, and arctic operations—the need for innovative approaches to flow assurance becomes increasingly critical (Bai & Bai, 2019; Prakoso et al. 2018; Punase et al. 2019; Kelland, 2014).

Paraffin wax deposition represents one of the most persistent and costly challenges in oil production and transportation systems. As crude oil flows from high-temperature reservoir conditions to cooler surface environments, the temperature drop triggers the precipitation of high molecular weight n-paraffins when reaching the wax appearance temperature (WAT). This precipitation initiates a complex crystallization process that can lead to significant operational issues throughout the production chain (Huang et al., 2016; Venkatesan et al., 2019). The complexity of paraffin wax management is compounded by the variable composition of crude oils across different fields and even within the same reservoir over time (Paso et al., 2009). Each crude oil contains a unique distribution of n-paraffins with different chain lengths and crystallization behaviors, making it difficult to develop universal mitigation strategies (Ganeeva et al., 2016). Furthermore, the dynamic nature of production operations introduces additional variables that affect wax deposition rates, including flow regimes, shear rates, pressure fluctuations, and the presence of other components like asphaltenes, resins, and inorganic solids that can serve as nucleation sites for wax crystal growth (Mehrotra & Bhat, 2007; Theyab, 2018; Punase et al. 2023).

Despite decades of research and application, significant technological gaps persist in the development of effective inhibitors for flow assurance. A fundamental challenge remains the limited molecular-level understanding of inhibitor interactions across diverse crude oil compositions. While the general mechanisms of crystal modification and co-crystallization are well established for paraffin inhibitors, the specific molecular features that determine inhibitor efficacy for different wax distributions are not fully characterized (Yang et al., 2015). This knowledge gap has resulted in an empirical approach to inhibitor development that relies heavily on trial-and-error testing rather than predictive design principles. The performance limitations of conventional polymeric inhibitors represent another significant challenge. Traditional pour point depressants (PPDs), including ethylene-vinyl acetate (EVA) copolymers, polyethylene-poly(ethylene-propylene) copolymers, and various methacrylate-based polymers, often show reduced effectiveness at low temperatures, in high-wax content crudes, or in the presence of other components like asphaltenes and resins (Machado et al., 2001; Wei, 2015; Aguiar et al. 2019a; Aguiar et al 2019b). Additionally, these inhibitors typically require relatively high dosages (500-2000 ppm) to achieve adequate performance, increasing operational costs (Kelland, 2014).

Manually evaluating multi-component formulations to determine synergistic effects and optimal blending ratios is extremely timeconsuming. A potential solution lies in automated systems capable of both formulating and testing a broad range of chemical combinations to identify the most effective product for each specific oil type. Combinatorial chemistry, a technique for rapidly generating diverse chemical compounds using automated methods and screening for desired properties, has existed for over five decades

and has been particularly successful in pharmaceutical research and drug discovery. In theory, millions of variants can be quickly produced and evaluated through automation. However, identifying the composition of successful "hits" often proves challenging, as does managing the extensive data generated (Hogan Jr., 1997; Studt, 1997).

It has become apparent that randomly combining initial variables for combinatorial synthesis and high-throughput screening is inefficient. The recursive deconvolution strategies used to identify effective variants may not succeed under these conditions. A more logical approach involves reducing the number of variants from the beginning. Rather than using robotics to rapidly blend numerous formulations, Design of Experiment (DoE) methodology can strategically limit the number of chemical combinations (Wu & Hamada, 2000; Anderson & Whitcomb, 2000; Montgomery, 2001). DoE applies geometric principles to statistical sampling to achieve specific goals, such as minimizing necessary experiments or reducing variance in regression-estimated coefficients. This approach can transform thousands of potential variants into a more manageable and meaningful set, enabling more efficient development of effective inhibitor formulations.

Recent advances in high-throughput technologies have transformed the development of flow assurance chemicals. Early work by Jennings and Weispfennig (2005) introduced miniaturized cold finger arrays, enabling parallel evaluation of up to 24 wax inhibitor formulations under identical conditions, while drastically reducing sample requirements from 500 mL to 20 mL per test. Building on this concept, Coutinho et al. (2018) developed a micro-differential scanning calorimetry (g-DSC) array capable of simultaneously analyzing 96 samples, measuring wax appearance temperatures and crystallization enthalpies, thereby allowing rapid assessment of inhibitor performance across concentration gradients. Parallel polymer synthesis has further accelerated inhibitor discovery. Guo et al. (2016) generated libraries of methacrylate-based pour point depressants with systematic variations in chain length, branching, and functional group density, combining automated synthesis with high-throughput rheological characterization to establish structure-property relationships. Similarly, Machado and Lucas (2011) employed combinatorial synthesis to produce and screen over 200 ethylene-vinyl acetate copolymer variants, revealing clear correlations between polymer architecture and wax inhibition efficiency.

Automation has also dramatically enhanced formulation and testing efficiency. Kelland et al. (2009) reported a robotic platform capable of preparing and evaluating up to 100 wax inhibitor formulations daily, integrating automated pour point measurements, cold filtration tests, and microscopic crystal analyses. Subramanian et al. (2021) extended this approach by combining automated formulation with rheological and microscopic imaging, enabling simultaneous evaluation of both macroscopic flow properties and microscopic crystal modification effects. The integration of computational modeling with high-throughput experimentation represents a particularly promising strategy. Dunn et al. (2019) demonstrated how molecular modeling can guide the selection of promising inhibitor structures, which are subsequently synthesized and tested in parallel arrays. This hybrid approach not only reduces the number of compounds requiring experimental evaluation but also expands the chemical space explored, accelerating the discovery of highly effective wax inhibitors.

Offshore and deepwater production environments present particularly severe challenges for flow assurance that can benefit from these advanced development approaches. The extreme temperature gradients encountered as hot crude oil travels through subsea flowlines exposed to near-freezing seawater temperatures create ideal conditions for rapid wax precipitation and asphaltene destabilization (Azevedo & Teixeira, 2003). The remote nature of these facilities makes remediation efforts logistically complex and extraordinarily expensive, with intervention costs potentially reaching millions of dollars per incident (Merino-Garcia & Corraera, 2008). Additionally, the increasing trend toward longer tiebacks and subsea processing further exacerbates these challenges by extending the residence time of oil in cold environments.

Environmental regulations and sustainability concerns are simultaneously driving significant innovation in green chemistry approaches to flow assurance. The development of bio-based alternatives derived from

renewable resources represents a promising direction, with recent studies demonstrating the effectiveness of modified natural polymers, including cellulose derivatives, chitosan, and plant-derived compounds as wax crystal modifiers (Norland, 2016; Sivakumar et al., 2018). These bio-based inhibitors not only address environmental concerns but, in some cases, offer performance advantages through unique structural features not easily replicated in synthetic polymers (Anisuzzaman et al., 2017).

Recent publications indicate growing interest in applying high throughput approaches to develop multi-functional additives that address multiple flow assurance challenges simultaneously. Song et al. (2020) described the use of parallel screening methods to evaluate nanoparticle-polymer hybrid materials as combined wax inhibitors and pour point depressants, while Ridzuan et al. (2020) reported high-throughput evaluation of inhibitor formulations for their effects on both wax and asphaltene stability. This trend towards integrated solutions reflects the recognition that flow assurance challenges rarely occur in isolation, and effective management strategies must consider the interplay between various deposition mechanisms.

The economic impact of flow assurance issues extends beyond immediate remediation costs to include production losses, increased energy consumption, accelerated equipment deterioration, and environmental risks associated with potential leaks or spills from compromised infrastructure (Bai & Bai, 2019). As industry moves toward more challenging production environments and aging infrastructure, the need for innovative, adaptive approaches to flow assurance becomes increasingly critical. High-throughput experimentation, combined with strategic design of experiments and computational modeling, offers a promising pathway to develop the next generation of flow assurance solutions capable of addressing the complex and evolving challenges of modern oil production systems. The integration of these advanced development methodologies with field testing and operational feedback loops represents the next frontier in flow assurance chemical development. By combining rapid laboratory screening with real-world performance data, researchers can continuously refine and optimize inhibitor formulations to meet the specific challenges of each production system. This holistic approach to flow assurance chemical development promises to deliver more effective, economical, and environmentally sustainable solutions for the petroleum industry's most persistent flow assurance challenges.

In this work, we build on these advancements by integrating DoE-driven formulation design with high-throughput experimental capabilities to develop next-generation paraffin inhibitors and pour point depressants. Our approach aims to bridge the gap between traditional trial-and-error chemistry and predictive, adaptive formulation strategies, delivering faster, more cost-effective, and environmentally responsible solutions to the complex flow assurance challenges faced in modern oil production.

Experimental Procedure

The process for High Throughput Experimentation (HTE) is based on the four main pillars of Smart Automation, Miniaturization, Parallelization, and Computational Design/Evaluation. Figure 1 outlines the importance and contribution of each of these pillars towards the overall process. Using a HTE workstation not only expedites the testing process but can also considerably reduce the crude oil volume requirement for testing hundreds of chemistries. The process can also reduce the error margin associated with a manual testing procedure and improve the overall system reproducibility and reliability. In addition to screening the potential chemistries at the initial life (start-up) of the well, the process has great utility in critical times when well characteristics may change unexpectedly to induce paraffin instability and imminent need of re-evaluation is necessary to manage the paraffin risk.

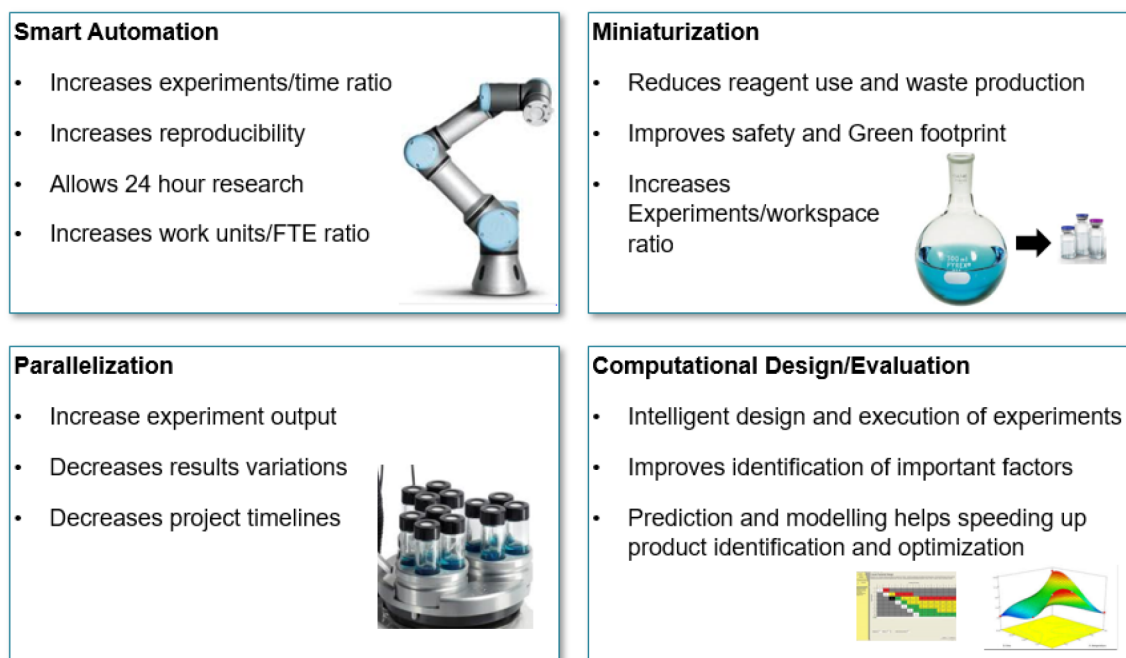


Figure 1—High Throughput Experimentation Workflow

Based on the principle of rheology, the HTE process measures the viscosities of the untreated and treated crude oil samples across a specified temperature range. Traditional cup and cone rheometer and cold finger methods were also used to conduct confirmatory tests on the selected formulations. In terms of oils, various oils from around the globe with paraffin associated challenges were evaluated in this study. Rheological studies were conducted at a temperature range of 60°C to -10°C with 1 to 10 sec⁻¹ shear rate. For cold finger tests, the oil samples were heated at 40°C and the finger temperatures were maintained at 4°C. The stirring rate was 300 rpm for all the tests, and the test was run for a 16-hour period before measuring the overall deposited mass.

Results and Discussion

The workflow for paraffin inhibitor / pour point depressant evaluation is outlined in Figure 2 and begins with the careful selection of base components and boosters. These represent the fundamental building blocks of PPD chemistry and provide the flexibility to explore a wide range of formulation strategies. To systematically organize this process, a design of experiments (DoE) approach was applied, allowing structured combinations of single bases, double bases, and base-booster mixes to be generated. This ensures that both individual effects and potential synergies between components are captured in the evaluation. Once the experimental plan was defined, a robotic formulation platform was used to prepare the mixtures. This automation step ensures precision and reproducibility, minimizing the variability often associated with manual handling. The outcome of this stage is a PPD toolkit, consisting of different formulation classes such as single base, double base, single base with booster, and double base with booster systems.

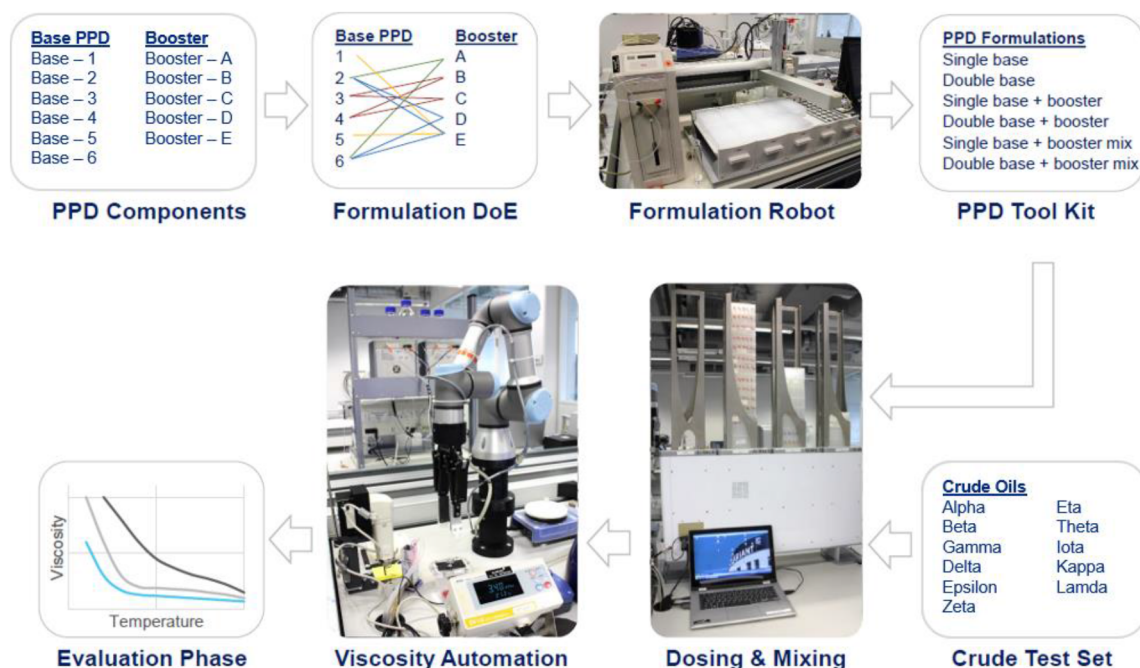


Figure 2—High Throughput Experimentation workflow for the paraffin inhibitor development study

The prepared formulations were then evaluated in a diverse set of crude oils, labeled Alpha to Lambda. This crude test set was chosen to represent the natural variability of crude oils and to provide a robust performance assessment across different compositions and rheological behaviors. Automated dosing and mixing equipment introduced the PPDs into the crude oils with precise concentration control, ensuring uniformity and consistency in sample preparation. Following sample preparation, viscosity automation systems measured the flow behavior of the crude-PPD blends. These robotic platforms capture viscosity-temperature profiles with high accuracy and throughput, providing the data needed to assess formulation effectiveness. In the final evaluation phase, the results were analyzed by comparing viscosity curves of untreated and treated crudes. This step highlights the impact of each formulation on pour point depression and flow improvement, ultimately guiding the selection of the most effective candidates for further development.

Figure 3 presents the viscosity profiles of all formulations tested on the Alpha crude oil at a fixed dosage of 500 ppm. Each curve corresponds to the performance of an individual product within the bulk fluid system. The yellow curve, which shows the highest viscosity values, represents the untreated Alpha crude oil and clearly demonstrates the effectiveness of all tested formulations in reducing viscosity. Similar trends were observed for all the other oil samples with the majority of the test formulations yielding a positive response to the bulk fluid viscosity reduction with respect to the untreated oil.

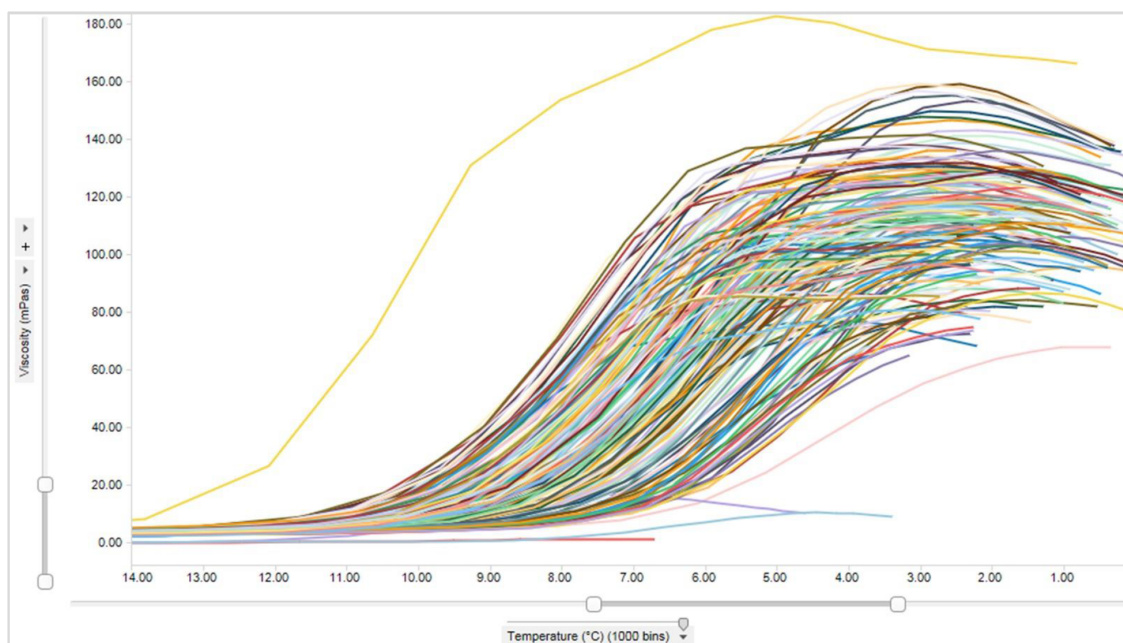


Figure 3—Viscosity profiles of the tested formulations at 500 ppm dosage on the Alpha oil

To ensure reliability, cross-validation of the high-throughput experimentation (HTE) results were carried out. This step is critical to detect and account for any potential false negatives that could otherwise be overlooked. For this purpose, five randomly selected formulations (50, 60, 86, 101, and 120) were further evaluated using a conventional rheometer. The comparison is illustrated in Figure 4, where red curves represent measurements from the conventional method and blue curves reflect the HTE data. The close agreement between the two sets of results confirms the accuracy of the HTE approach and reinforces confidence in its outcomes.

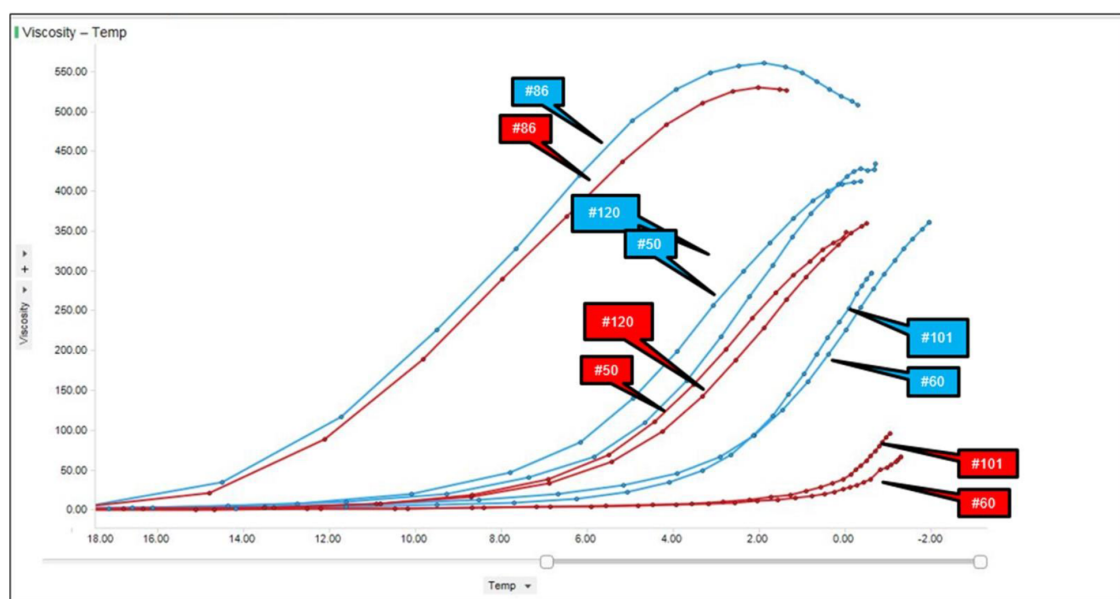


Figure 4—Cross validation of the HTE results with conventional viscometer measurements

Building upon the dataset generated, the ten best-performing formulations (RP-1 to RP-10) were selected for further testing on seven problematic crude oils sourced globally. These oils are known to cause operational issues such as fluid gelation (high pour points) or wax deposition in flowlines, transport

pipelines, and topside facilities. A conventional rheometer equipped with a cup-and-cone assembly was used to measure viscosity over a temperature range from -10°C to 60°C . For clarity, Figure 5 presents the maximum viscosity values recorded at -10°C , the lowest test temperature. Each bar in the graph represents either the benchmark or RP products applied at a dosage of 500 ppm in the different oil samples. A logarithmic scale was employed to account for the very high viscosity of certain samples, particularly Oil-7, a heavy crude from Latin America. The results indicate that not all RP formulations outperformed the benchmark, and their relative effectiveness varied across the oils. This inconsistency highlights the importance of developing more tailored inhibitor solutions that align with the specific properties of each crude oil.

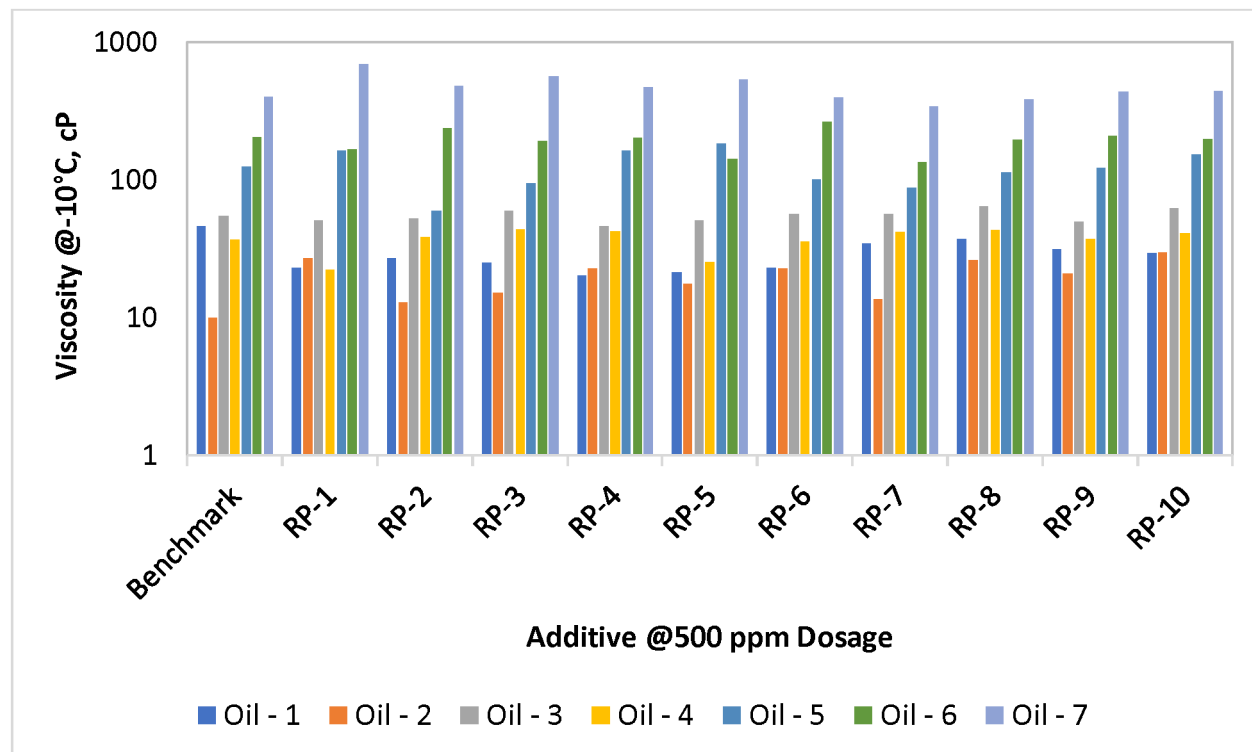


Figure 5—Viscosity results at -10°C for seven crude oil samples with benchmark and ten RP products at 500 ppm dosage

Paraffin inhibitor performance can differ significantly depending on whether it is evaluated using *cold finger* tests, which measure surface deposition control, or *rheology* tests, which assess bulk flow property modification. These differences are driven by variations in chemical structure, test conditions, crude oil composition, dosage effects, and operational factors.

From a chemical structure standpoint, medium-to-long polymer chains with higher polarity often excel in cold finger tests by disrupting wax crystal lattice formation on surfaces, while shorter-to-medium chains with balanced polar/non-polar distribution typically perform better in rheology tests by improving bulk flow properties. Co-polymer composition can also create trade-offs, with some formulations effective for deposition prevention but having less impact on flow behavior. Test conditions can further influence results. Cold finger performance is highly dependent on temperature gradients, cold surface temperature, and low-shear environments, whereas rheology tests are more sensitive to thermal history, shear rate, and temperature cycling. Differences also emerge from crude oil composition—cold finger tests are more influenced by high molecular weight paraffins (C_{30+}), while rheology tests respond more to mid-range paraffins ($\text{C}_{18}\text{-C}_{30}$). Asphaltenes, resins, and solvent interactions can enhance or hinder performance in method-specific ways. The impact of concentration can differ across testing methods; cold finger evaluations typically demonstrate a proportional relationship with dosage amounts, whereas rheological measurements may exhibit critical

thresholds or stabilized response regions. Additional considerations such as synergistic blends, critical micelle concentration effects, and aging-related performance loss underscore the importance of method-specific behavior. Because each method highlights different mechanisms—surface deposition control versus bulk flow modification—effective inhibitor selection should rely on a combination of test data that reflects actual field conditions and operational priorities.

Accordingly, the RP products were also tested on cold finger for each of the seven oil samples. For the deposition based cold finger test, the additive dosage was increased to 1000 ppm to gain more visibility of the performance. Figure 6 shows the overall paraffin mass deposit on the finger. Similar to the rheological data, cold finger results also highlight the performance variation of RP products across oil samples. However, it should be noted that there were more positive hits of RP products outperforming the benchmark formulation on the cold finger test when compared to the rheological study.

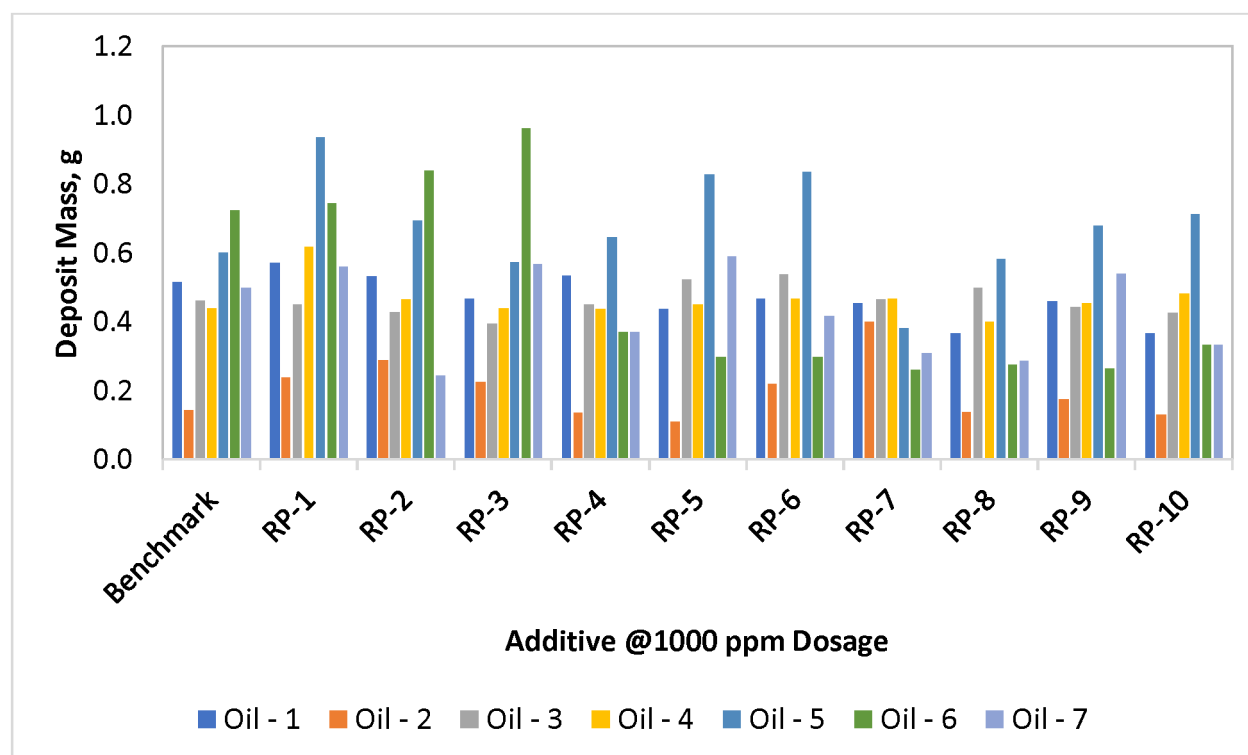


Figure 6—Cold finger results for seven crude oil samples with benchmark and ten RP products at 1000 ppm dosage

To further evaluate the relationship between bulk rheological behavior and surface deposition tendencies, the cold finger deposit mass was plotted against the maximum viscosity values for each of the seven crude oils (Figure 7). The resulting regression fits highlight oil-specific trends with varying degrees of correlation (R^2 values). For Oils 5 and 7, strong positive correlations were observed ($R^2 = 0.8732$ and 0.7769 , respectively), indicating that higher viscosity responses are closely associated with increased paraffin deposition. This suggests that, for these oils, the mechanisms governing bulk crystallization and surface deposition are strongly coupled, and rheological data may serve as a reliable predictor of deposition behavior. Oil 2 also showed a moderate correlation ($R^2 = 0.6401$), reinforcing a partial alignment between the two test methods. In contrast, Oils 1, 3, 4, and 6 exhibited negligible correlations ($R^2 < 0.1$). For these crude oils, cold finger results could not be predicted from viscosity measurements, highlighting a decoupling between surface and bulk phenomena. These discrepancies likely stem from compositional differences—such as varying paraffin carbon number distributions, asphaltene/resin content, and solvent-solute interactions—that uniquely influence deposition and gelation mechanisms.

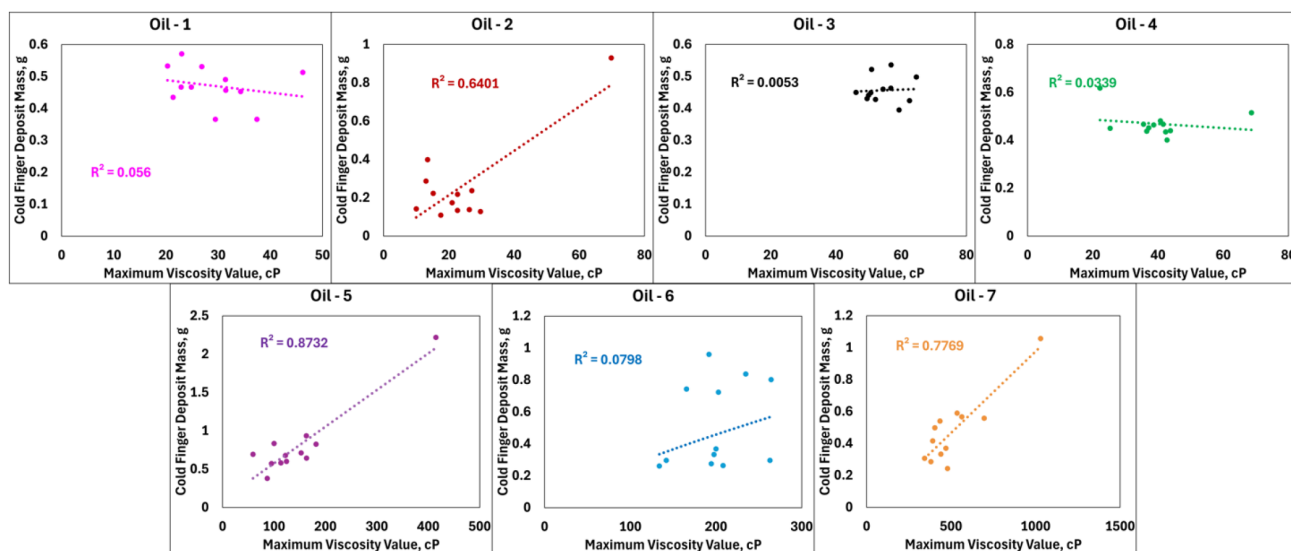


Figure 7—Correlation of bulk viscosity and cold finger results for the seven crude oil samples

Overall, these findings emphasize that the predictive value of rheological data for deposition control is highly crude-dependent. While strong correlations in some oils indicate shared mechanisms, the lack of alignment in others underscores the importance of using both cold finger and rheology tests in parallel. A combined evaluation provides a more comprehensive understanding of inhibitor performance by capturing both surface deposition control and bulk flow modification, ensuring more robust candidate selection under field-relevant conditions.

Another way of clearly depicting the rheology and cold finger results is to denote the RP products outperforming benchmark formulation as ‘R’ (in Table - 1) or ‘D’ (in Table - 2) for each test. From Table - 1, we can observe that inhibitor performance is highly oil- dependent, with no single RP product consistently effective across all crude oils. RP-7 and RP-8 demonstrated the broadest spectrum of activity, each improving rheological behavior in four oils, including the particularly challenging Oils 6 and 7. RP-3 and RP-5 also showed favorable performance but with narrower applicability. In contrast, Oil-2 and Oil-7 exhibited limited responsiveness overall, as most RP formulations showed no improvement relative to the benchmark.

Table 1—Rheological assessment of the seven crude oil samples with respect to the benchmark product

Rheological Evaluation							
Sample	Oil - 1	Oil - 2	Oil - 3	Oil - 4	Oil - 5	Oil - 6	Oil - 7
Benchmark	0	0	0	0	0	0	0
RP-1	R	0	R	R	0	R	0
RP-2	R	0	R	0	R	0	0
RP-3	R	0	0	0	R	R	0
RP-4	R	0	R	0	0	R	0
RP-5	R	0	R	R	0	R	0
RP-6	R	0	0	R	R	0	R
RP-7	R	0	0	0	R	R	R
RP-8	R	0	0	0	R	R	R
RP-9	R	0	R	0	R	0	0
RP-10	R	0	0	0	0	R	0

Table 2—Cold finger deposition assessment of the seven crude oil samples with respect to the benchmark product

Deposition Evaluation							
Sample	Oil - 1	Oil - 2	Oil - 3	Oil - 4	Oil - 5	Oil - 6	Oil - 7
Benchmark	0	0	0	0	0	0	0
RP-1	0	0	D	0	0	0	0
RP-2	0	0	D	0	0	0	D
RP-3	D	0	D	0	D	0	0
RP-4	0	D	D	D	0	D	D
RP-5	D	D	0	0	0	D	0
RP-6	D	0	0	0	0	D	D
RP-7	D	0	0	0	D	D	D
RP-8	D	D	0	D	D	D	D
RP-9	D	0	D	0	0	D	0
RP-10	D	D	D	0	0	D	D

The results from Table - 2 indicate that a larger number of RP products outperformed the benchmark in deposition control compared to rheological improvement, confirming that cold finger testing captures complementary performance aspects. RP-8 demonstrated the broadest efficacy, reducing deposition in six out of seven oils, including the challenging Oils 6 and 7. RP-10 also showed broad applicability with improvements in five oils. RP-4 and RP-7 displayed activity across four oils each, with RP-7 being notable for its performance in Oils 5, 6, and 7 where rheological challenges were most severe. On the other hand, Oils 1 and 2 showed fewer positive responses overall, with most RP products yielding little to no improvement. Oil-7 proved to be particularly resistant, with only RP-6, RP-7, RP-8, and RP-10 showing measurable deposition control. Taken together, these deposition evaluations reinforce the value of integrating both rheological and deposition-based assessments. While rheology highlights inhibitors that improve bulk flow, cold finger testing identifies candidates that mitigate surface-driven wax deposition. Importantly, formulations such as RP-7 and RP-8 emerge as strong multi-oil candidates with dual benefits, making them promising for further development under field-representative conditions.

The combination evaluation highlights the limited number of formulations capable of delivering simultaneous improvements in both rheology and deposition control. Only a subset of RP products demonstrated such dual effectiveness, and their impact was highly oil specific. Among these, RP-7 and RP-8 emerged as the most promising candidates, showing combination benefits across three oils, including the particularly challenging Oils 5, 6, and 7. Other formulations such as RP-3, RP-4, RP-5, RP-6, and RP-10 provided dual improvements in select oils but lacked broad applicability, while RP-1, RP-2, and RP-9 showed combination effects in only isolated cases. These findings underscore that while many inhibitors show value in one domain, only a few can consistently influence both bulk flow and deposition, reinforcing the need to prioritize dual-acting candidates such as RP-7 and RP-8 for further evaluation.

Table 3—Combined evaluation of the performance efficiency of ten RP products against the benchmark product on seven crude oil samples

Combination Evaluation							
Sample	Oil - 1	Oil - 2	Oil - 3	Oil - 4	Oil - 5	Oil - 6	Oil - 7
Benchmark	0	0	0	0	0	0	0
RP-1	0	0	C	0	0	0	0
RP-2	0	0	C	0	0	0	0
RP-3	C	0	0	0	C	0	0
RP-4	0	0	C	0	0	C	0
RP-5	C	0	0	0	0	C	0
RP-6	C	0	0	0	0	0	C
RP-7	C	0	0	0	C	C	C
RP-8	C	0	0	0	C	C	C
RP-9	C	0	C	0	0	0	0
RP-10	C	0	0	0	0	C	0

Conclusions

The high-throughput experimentation workflow established in this study proved to be an effective approach for screening and evaluating paraffin inhibitors and pour point depressant formulations across a diverse set of crude oils. By integrating systematic formulation design, robotic preparation, and automated viscosity measurements, the methodology allowed rapid and reproducible identification of high-performing candidates while minimizing manual variability. Cross-validation with conventional rheology confirmed the reliability of the HTE results, demonstrating strong agreement for selected formulations and reinforcing the robustness of the screening process.

Performance assessments across seven challenging crude oils revealed that inhibitor efficacy is highly oil-dependent, with no single formulation proving to be universally effective. While many RP products improved either bulk rheology or surface deposition control, only a subset—most notably RP-7 and RP-8—exhibited dual functionality across multiple oils, making them promising candidates for further development. The study also highlighted that correlations between rheological improvement and cold finger deposition control are crude-specific, emphasizing the importance of combining both test approaches to fully capture inhibitor performance. Overall, this integrated evaluation framework provides a comprehensive and efficient strategy for selecting paraffin inhibitors capable of addressing both flow assurance and deposition challenges in field-relevant conditions.

References

- Aguiar, J. I., Pimentel Porto Mazzeo, C., Garan, R., Punase, A., Razavi, S., Mahmoudkhani, A. 2019a. What Can We Learn from Analysis of Field Asphaltenes Deposits? Enhancing Product Development Through Knowledge-Based Field-to-Lab-to-Field Approach. Presented at SPE International Conference on Oilfield Chemistry, Galveston, USA. SPE-193604-MS.
- Aguiar, J. I., Punase, A., & Mazzeo, C. 2019b. Influence of Asphaltene Inhibitors on Wax and Asphaltene Deposition - Are Problems Associated?. Presented at Offshore Technology Conference Brasil, Brazil, OTC-29817-MS.
- Akbarzadeh, K., Ratulowski, J., Lindvig, T., Davies, T., Huo, Z., Broze, G., & Hemmingsen, P. V. 2020. The Application of Machine Learning Algorithms in Developing Wax Deposition Mitigation Strategies. *Energy & Fuels*, **34**(6), 6796–6808.
- Anderson, M. J., & Whitcomb, P. J. 2000. DOE Simplified: Practical Tools for Effective Experimentation. *Productivity Press*.
- Anisuzzaman, S.M., Abang, S., Bono, A., Krishnaiah, D., Karali, R., Safuan, M.K. 2017. Wax Inhibitor based on Ethylene Vinyl Acetate with Methyl Methacrylate and Diethanolamine for Crude Oil Pipeline. *IOP Conference Series: Materials Science and Engineering*, **206**(1), 012074.
- Azevedo, L. F. A., Teixeira, A. M. 2003. A Critical Review of the Modeling of Wax Deposition Mechanisms. *Petroleum Science and Technology*, **21**(3-4), 393–408.

- Bai, Y., Bai, Q. 2019. *Subsea Engineering Handbook*. Gulf Professional Publishing.
- Chi, Y., Daraboina, N., Sarica, C. 2019. Effect of the Flow Field on the Wax Deposition and Performance of Wax Inhibitors: Cold Finger and Flow Loop Testing. *Energy & Fuels*, **33**(3), 2019–2033.
- Coutinho, J. A., Lopes da Silva, J.A., Ferreira, A., Soares, M.R., Daridon, J.L. 2018. Evidence for the Aging of Wax Deposits in Crude Oils by Ostwald Ripening. *Petroleum Science and Technology*, **36**(13), 1045–1051.
- Dunn, P. J., Wells, A. S., Williams, M. T. (Eds.). 2019. *Green Chemistry in the Pharmaceutical Industry*. John Wiley & Sons.
- Ganeeva, Y. M., Yusupova, T. N., Romanov, G. V. 2016. Waxes in Asphaltenes of Crude Oils and Wax Deposits. *Petroleum Science*, **13**(4), 737–745.
- Gao, S., Moran, K., Xu, Z., Masliyah, J. 2018. Role of Nanoparticles in Oil Sand Processing. *Energy & Fuels*, **32**(3), 3338–3350.
- Guo, X., Pethica, B. A., Huang, J. S., Adamson, D. H., Prud'homme, R.K. 2016. Effect of Cooling Rate on Crystallization of Model Waxy Oils with Microcrystalline Poly(ethylene butene). *Energy & Fuels*, **20**(1), 250–256.
- Hogan, C.L. Jr. 1997. Combinatorial Chemistry in Materials Science. *Materials Science and Engineering: R: Reports*, **20**(3), 105–200.
- Huang, Z., Zheng, S., Fogler, H.S. 2016. *Wax Deposition: Experimental Characterizations, Theoretical Modeling, and Field Applications*. CRC Press.
- Jafari Ansaroudi, H. R., Vafaie-Sefti, M., Masoudi, S., Behbahani, T. J., Jafari, H. 2018. Experimental Study and Modeling of Wax Deposition in the Presence and Absence of the Pour Point Depressant and Wax Inhibitor. *Brazilian Journal of Chemical Engineering*, **35**(2), 399–412.
- Jennings, D. W., Weispfennig, K. 2005. Effects of Shear and Temperature on Wax Deposition: Cold Finger Investigation with a Gulf of Mexico Crude Oil. *Energy & Fuels*, **19**(4), 1376–1386.
- Kang, P. S., Lee, D. G., Lim, J. S. 2014. Status of Wax Mitigation Technologies in Offshore Oil Production. *Journal of Industrial and Engineering Chemistry*, **20**(4), 1–12.
- Kelland, M. A. 2014. *Production Chemicals for the Oil and Gas Industry* (2nd ed.). CRC Press.
- Kelland, M. A., Leporcher, E., Lohne, A., Boe, E. 2009. Systematic Development and Verification of Wax Inhibitor Test Methods. Presented at SPE International Symposium on Oilfield Chemistry, The Woodlands, Texas.
- Machado, A. L. C., Lucas, E. F. 2011. Influence of Ethylene-co-vinyl Acetate Copolymers on the Flow Properties of Wax Synthetic Systems. *Journal of Applied Polymer Science*, **122**(3), 1905–1912.
- Machado, A. L. C., Lucas, E. F., Gonzalez, G. 2001. Poly(ethylene-co-vinyl acetate) (EVA) as Wax Inhibitor of a Brazilian Crude Oil: Oil Viscosity, Pour Point and Phase Behavior of Organic Solutions. *Journal of Petroleum Science and Engineering*, **32**(2-4), 159–165.
- Mehrotra, A. K., Bhat, N. V. 2007. Modeling the Effect of Shear Stress on Deposition from "Waxy" Mixtures under Laminar flow with Heat Transfer. *Energy & Fuels*, **21**(3), 1277–1286.
- Merino-Garcia, D., Corraera, S. 2008. Cold Flow: A Review of Technology to Avoid Wax Deposition. *Petroleum Science and Technology*, **26**(4), 446–459.
- Montgomery, D. C. 2001. *Design and Analysis of Experiments* (5th ed.). John Wiley & Sons.
- Norland, A. K. 2016. Bio-Based Chemicals for Flow Assurance. Presented in SPE International Oilfield Chemistry Conference, Montgomery, Texas.
- Paso, K., Senra, M., Yi, Y., Sastry, A. M., Fogler, H. S. 2009. Paraffin Polydispersity Facilitates Mechanical Gelation. *Industrial & Engineering Chemistry Research*, **48**(16), 7660–7668.
- Prakoso, A., Punase, A., Rogel, E., Ovalles, C., Hascakir, B. 2018. Effect of Asphaltene Characteristics on its Solubility and Overall Stability. *Energy & Fuels*, **32**(6), 6482–6487.
- Punase, A., Aguiar, J., Smith, R., Mazzeo, C., Wylde, J. 2019. Importance of Comprehending Waxphaltenes and the Need to Develop New Inhibitors to Prevent its Deposition. Paper presented at the SPE/IATMI Asia Pacific Oil & Gas Conference and Exhibition, Bali, Indonesia, SPE-196368-MS.
- Punase, A., Smith, R., Burnett, W., & Mutch, K. 2023. Impact of Waxphaltenes on Overall Stability of Asphaltenes, Paraffins, and Emulsion. Paper presented at the SPE Abu Dhabi International Petroleum Exhibition and Conference, UAE, SPE-216472-MS.
- Ridzuan, N., Adam, F., Yaacob, Z. 2020. Molecular Recognition of Wax Inhibitor Through Chemical Functionality and Performance Testing. *Petroleum Science*, **17**, 1000–1010.
- Sivakumar, P., Sircar, A., Deka, B., Anumegalai, A. S., Moorthi, P. S., Yasvanthrajan, N. 2018. Flow Improvers for Assured Flow of Crude Oil in Midstream Pipeline - A Review. *Journal of Petroleum Science and Engineering*, **164**, 24–30.
- Song, X., Yin, H., Feng, Y., Zhang, S., Wang, Y. 2020. Effect Of SiO₂ Nanoparticles on Wax Crystallization and Flow Behavior of Model Crude Oil. *Industrial & Engineering Chemistry Research*, **59**(22), 10481–10490.
- Sousa, A. L., Matos, H. A., Guerreiro, L. P. 2019. Preventing and Removing Wax Deposition Inside Vertical Wells: A Review. *Journal of Petroleum Exploration and Production Technology*, **9**(3), 2091–2107.

- Studt, T. 1997. Combinatorial Chemistry: Synthesizing a Solution to the Research Productivity Problem. *R&D Magazine*, **39**(12), 24–28.
- Subramanian, D., May, A. F., Katz, B. 2021. Responsive Polymer Systems for Controlled Delivery of Flow Assurance Chemicals. *Energy & Fuels*, **35**(7), 6271–6282.
- Theyab, M. A. 2018. Wax Deposition Process: Mechanisms, Affecting Factors and Mitigation Methods. *Open Access Journal of Science*, **2**(2), 112–118.
- Venkatesan, R., Ostlund, J. A., Chawla, H., Wattana, P., Nyden, M., Fogler, H. S. 2019. The Effect of Asphaltenes on the Gelation of Waxy Oils. *Energy & Fuels*, **17**(6), 1630–1640.
- Wei, B. 2015. Recent Advances on Mitigating Wax Problem Using Polymeric Wax Crystal Modifier. *Journal of Petroleum Exploration and Production Technology*, **5**(4), 391–401.
- Wu, C. F. J., Hamada, M. 2000. *Experiments: Planning, Analysis, And Parameter Design Optimization*. John Wiley & Sons.
- Yang, F., Zhao, Y., Sjoblom, J., Li, C., Paso, K. G. 2015. Polymeric Wax Inhibitors and Pour Point Depressants for Waxy Crude Oils: A Critical Review. *Journal of Dispersion Science and Technology*, **36**(2), 213–225.
- Zhu, T., Walker, J., Liang, J. 2021. Application of Machine Learning in Flow Assurance: Wax Deposition Prediction and Inhibitor Selection. *SPE Production & Operations*, **36**(02), 302–311.